

ATPL EASA STUDY GUIDE | 070 - OPERATIONAL PROCEDURES

Chapter 10: Carriage of Dangerous Goods by Air (ICAO & IATA)

Syllabus Source: CAE Oxford (pp. 347-380) | Pass Target: >= 90% AviationExam

1. THE 9 CLASSES OF DANGEROUS GOODS

Under ICAO Annex 18 and the IATA Dangerous Goods Regulations (DGR), hazardous materials are categorized into 9 distinct hazard classes:

Class	Dangerous Goods Category	Examples & Aircraft Loading Restrictions
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Class 1	Explosives (Divisions 1.1 to 1.6)	Ammunition, flares, fireworks. Most are forbidden on passenger aircraft.
Class 2	Gases (Flammable, Non-flammable, Toxic)	Aerosols, oxygen cylinders, compressed butane, toxic chlorine.
Class 3	Flammable Liquids	Paints, aviation fuels, alcohol solvents (flash point <= 60°C).
Class 4	Flammable Solids	Matches, white phosphorus, magnesium alloys, spontaneously combustible goods.
Class 5	Oxidizing Substances & Organic Peroxides	Chemical oxygen generators (ValuJet tragedy), hydrogen peroxide.
Class 6	Toxic & Infectious Substances	Pesticides, cyanide, medical viral samples (Division 6.2).
Class 7	Radioactive Material	Medical isotopes, industrial radiotracers. Classified into Category I, II, III.
Class 8	Corrosives	Sulfuric acid, battery fluids, mercury (corrodes aluminum alloy airframes!).
Class 9	Miscellaneous Dangerous Goods	LITHIUM-ION BATTERIES, dry ice (solid CO2 creates asphyxiation risk in holds).

2. NOTIFICATION TO CAPTAIN (NOTOC)

Before departure, the airline ground handling agent must deliver a written NOTOC document to the aircraft Commander:

- Mandatory NOTOC Data: 1. Proper Shipping Name and UN Number; 2. Hazard Class and Division; 3. Net quantity per package; 4. Exact physical cargo hold compartment location (e.g. Hold 4, Bin 2); 5. Confirmation that packages are undamaged; 6. ICAO Emergency Response Drill Code (e.g. Drill 4L for emergency checklists).

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): NOTOC Verification & In-Flight D

Notification to Captain (NOTOC) Verification: Step 1: Confirm written NOTOC is delivered before door closure: verify UN numbers, class, quantity, and exact cargo hold/bay. Step 2: Check ICAO Emergency Drill Code on NOTOC (e.g. 4L). Step 3: Cargo Fire / Spillage Action: - Discharge cargo fire extinguisher bottle 1 immediately (Halon). - Confirm slow-rate discharge bottle 2 initiates. - Divert to nearest suitable airport, declare MAYDAY, and pass UN numbers and drill codes to ATC.

3. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: What is the formal document that MUST be provided in writing to the aircraft Commander prior to departure?**

- [A]: Flight manifest
- [B]: NOTOC (Notification to Captain)
- [C]: Cargo bill
- [D]: Customs clearance

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Under ICAO Annex 18 and EASA SPA.DG.105, the Commander must receive a written NOTOC detailing all hazardous goods, positions, and emergency drill codes.

Q2: Under which IATA Dangerous Goods Class are LITHIUM-ION BATTERIES categorized?

- [A]: Class 1
- [B]: CLASS 9 (Miscellaneous Dangerous Goods)
- [C]: Class 3
- [D]: Class 5

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Lithium batteries (UN 3480 and UN 3481) fall under Class 9: Miscellaneous Dangerous Goods.

Q3: What hazard is associated with carrying large quantities of DRY ICE (solid carbon dioxide) in an aircraft cabin?

- [A]: Explosion risk
- [B]: SUFFOCATION / ASPHYXIATION HAZARD, as dry ice sublimates into gaseous carbon dioxide, displacing oxygen in confined compartments
- [C]: Radiation
- [D]: Corrosion of steel

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Dry ice sublimates into gaseous CO₂ at normal hold temperatures, creating an invisible asphyxiation risk for ground and flight personnel.